

Lakshya JEE 2.0 2025

Maths

DPP: 1

Method of Differentiation

Q1 If $y = \frac{1+x^2+x^4}{1+x+x^2}$ and $\frac{dy}{dx} = ax + b$; find $a + b$.

- (A) 0 (B) -1
(C) 1 (D) 2

Q2 If $y = \frac{x^2+2^x}{e^x}$ and $\frac{dy}{dx} = \frac{x(2p+qx)+2^x(\ln 2+r)}{e^x}$, find

$p + qr$

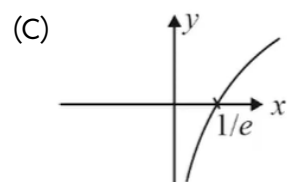
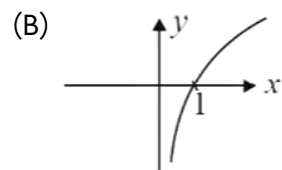
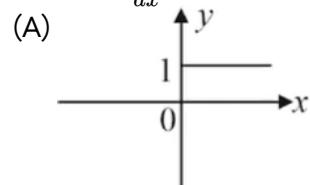
- (A) -2 (B) 2
(C) 0 (D) 1

Q3 If $y = (1 + x^{1/4})(1 + x^{1/2})(1 - x^{1/4})$, find

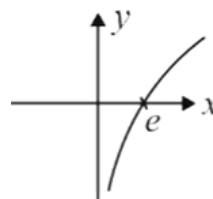
$\frac{dy}{dx}$

- (A) -2 (B) 1
(C) 0 (D) -1

Q4 Which of the following could be the sketch of graph of $\frac{d}{dx}(x \log x)$?



(D)



Q5 Derivative of $y = e^{(\sin x)}$ using first principle.

- (A) $e^{\cos x} \cdot \sin x$
(B) $e^{\sin x} \cdot \cos x$
(C) $e^{-\cos x} \cdot \sin x$
(D) $e^{-\sin x} \cdot \cos x$

Q6 If $y = \sqrt{x} + x\sqrt{x} + x^2\sqrt{x} + x^3\sqrt{x}$, find $\frac{dy}{dx}$

- (A) $\frac{1}{2\sqrt{x}} + \frac{3}{2}\sqrt{x} + \frac{5}{2}x^{1/2} + \frac{7}{2}x^{5/2}$
(B) $\frac{1}{2\sqrt{x}} + \frac{3}{2\sqrt{x}} + \frac{5}{2}x^{3/2} + \frac{7}{2}x^{5/2}$
(C) $\frac{1}{2\sqrt{x}} + \frac{3}{2}\sqrt{x} + \frac{5}{2}x^{3/2} + \frac{7}{2}x^{5/2}$
(D) $\frac{1}{2\sqrt{x}} + \frac{3}{2}\sqrt{x} + \frac{5}{2}x^{1/2} + \frac{7}{2}x^{3/2}$

Q7 If $y = \log(\sin x + \cos x)$, find $\frac{dy}{dx}$

- (A) $\cot\left(\frac{\pi}{4} - x\right)$
(B) $\cot\left(\frac{\pi}{4} + x\right)$
(C) $\tan\left(\frac{\pi}{4} + x\right)$
(D) $\tan\left(\frac{\pi}{4} - x\right)$

Q8 If $y = \log_{\sin x}(\tan x)$, then $\left(\frac{dy}{dx}\right)_{\pi/4}$ is equal

to:

- (A) $\frac{4}{\log 2}$
(B) $-4 \log 2$
(C) $\frac{-4}{\log 2}$
(D) None of these

Q9



Let $f(x) = \begin{vmatrix} 1 & 2x & 3x^2 \\ x & x^2 & x^3 \\ 0 & 2 & 6x \end{vmatrix}$ Then find the value

of $f'(1)$

- (A) 0 (B) -2
(C) -6 (D) 6

Q10 Find $\frac{dy}{dx}$, if $y = x^{\sin x}$.

- (A) $x^{\sin x} \left(\frac{\sin x}{x} - \cos x \log x \right)$
(B) $x^{\sin x} \left(\frac{\sin x}{x} + \cos x \log x \right)$
(C) $x^{\sin x} \left(\frac{\sin x}{x} + \cos x \cdot \frac{1}{x} \right)$
(D) $x^{\sin x} \left(\frac{\sin x}{x} + \sin x \log x \right)$

Q11 If $x^m y^n = (x + y)^{m+n}$, find $\frac{dy}{dx}$

- (A) $-\frac{m}{n} \cdot \frac{y}{x}$
(B) $\frac{m}{n} \frac{y}{x}$
(C) $\frac{y}{x}$
(D) $-\frac{y}{x}$

Q12 If $y = \left(1 + \frac{1}{x}\right)^x + x^{\left(1 + \frac{1}{x}\right)}$, find $\frac{dy}{dx}$ at $x = 1$

- (A) $\log 2 - \frac{1}{2}$
(B) $\log 2 + 1$
(C) $\log 2 - 1$
(D) $\log 4 + 1$

Q13 If $x^y = y^x$, find $\frac{dy}{dx}$

- (A) $\frac{y}{x} \left(\frac{x \log y - y}{y \log x - x} \right)$
(B) $\frac{y}{x} \left(\frac{y \log y - x}{x \log x - y} \right)$
(C) $\frac{x}{y} \left(\frac{x \log y - y}{y \log x - x} \right)$
(D) $\frac{x}{y} \left(\frac{y \log y - x}{x \log y - y} \right)$

Q14 If

$y = (1+x)(1+x^2)(1+x^4)(1+x^8) \dots (1+x^{1006})$
the value of $y'(0)$ is:

- (A) 503 (B) 0
(C) 1 (D) 1006

Q15 If $x^y = e^{x-y}$, then $\frac{dy}{dx} =$

- (A) $\frac{(\log x - 1)}{(1 + \log x)^2}$
(B) $\frac{\log x}{(1 + \log x)^2}$
(C) $\frac{1}{(1 + \log x)^2}$
(D) $-\frac{1}{(1 + \log x)^2}$

Q16 If $y(\alpha) = \sqrt{2 \left(\frac{\tan \alpha + \cot \alpha}{1 + \tan^2 \alpha} \right) + \frac{1}{\sin^2 \alpha}}$, α , then

$\in \left(\frac{3\pi}{4}, \pi \right)$

$\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is

- (A) 4 (B) 2
(C) 3 (D) -4



Answer Key

Q1 (C)

Q2 (B)

Q3 (D)

Q4 (C)

Q5 (B)

Q6 (C)

Q7 (D)

Q8 (C)

Q9 (C)

Q10 (B)

Q11 (C)

Q12 (D)

Q13 (A)

Q14 (C)

Q15 (B)

Q16 (A)

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